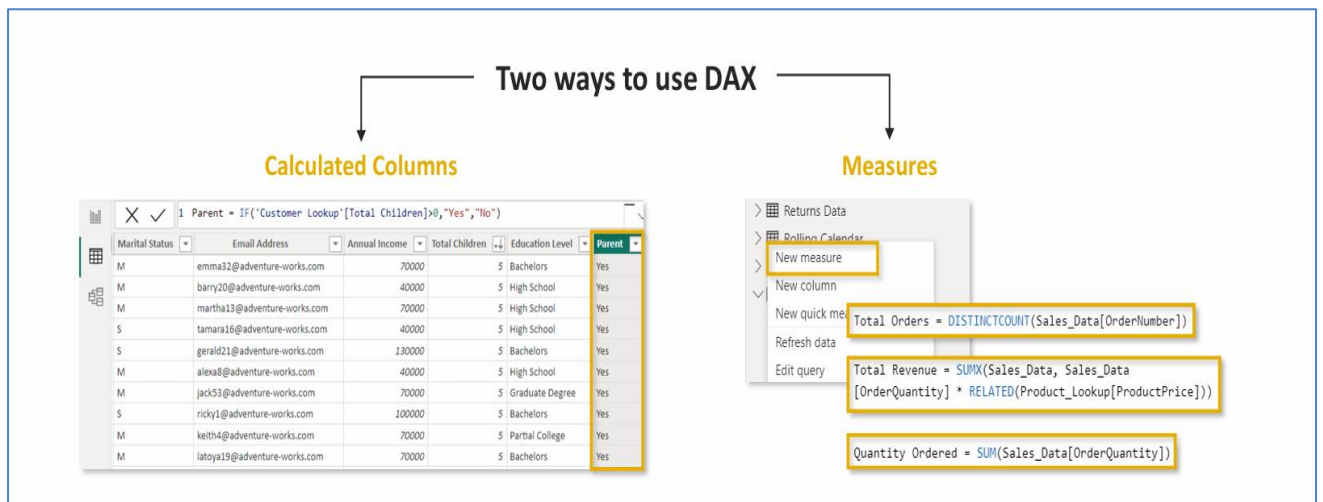


CALCULATED FIELDS WITH DAX

Data Analysis Expressions (commonly known as DAX) is the formula language that drives the Power BI front-end.

Add **calculated columns** (for filtering) and **measures** (for aggregation) to enhance data models



M VS. DAX

- **M** is used in the Power Query editor, and is designed specifically for extracting, transforming and loading data
- **DAX** is used in the Power BI front-end, and is designed specifically for analyzing relational data models

M

Query Editor:

PROPERTIES

Name: Territory Lookup

APPLIED STEPS

- Source
- Promoted Headers
- Changed Type**

```

#"Changed Type" = Table.TransformColumnTypes(
    #"Promoted Headers",
    {
        {"SalesTerritoryKey", Int64.Type},
        {"Region", type text},
        {"Country", type text},
        {"Continent", type text}
    }
)
    
```

// Adding a new step
// after we promoted headers
// that changes column datatypes

DAX

Report View:

Category Name	Total Returns	Bike Returns
Accessories	1,115	
Bikes	427	427
Clothing	267	
Total	1,809	427

```
1 Bike Returns =  
2 CALCULATE(  
3     [Total Returns],  
4     'Product Categories Lookup'[Category Name] = "Bikes"  
5 )
```

// Counting total returns
// filtered for bikes only

CALCULATED COLUMNS

- Calculated columns refer to entire tables or columns (no A1-style cell references)
- Calculated columns generate values for each row, which are visible within tables in the Data view
- Calculated columns understand row context; they're great for defining properties based on information in each row, but generally useless for aggregation (sum, count, etc.)
- **Calculated columns are typically used for filtering & grouping data, rather than creating aggregate numerical values**

EXAMPLE: CALCULATED COLUMNS

Formula Bar: `1 Parent = IF('Customer Lookup'[Total Children]>0,"Yes","No")`

Email Address	Annual Income	Total Children	Education Level	Parent
emma32@adventure-works.com	70000	5	Bachelors	Yes
barry20@adventure-works.com	40000	5	High School	Yes
martha13@adventure-works.com	70000	5	High School	Yes
tamara16@adventure-works.com	40000	5	High School	Yes
gerald21@adventure-works.com	130000	5	Bachelors	Yes
alexa8@adventure-works.com	40000	5	High School	Yes
jack53@adventure-works.com	70000	5	Graduate Degree	Yes
ricky1@adventure-works.com	100000	5	Bachelors	Yes
keith4@adventure-works.com	70000	5	Partial College	Yes
latoya19@adventure-works.com	70000	5	Bachelors	Yes

Fields List:

- Calendar Lookup
- Customer Lookup
 - Annual Income
 - Birth Date
 - Customer Key
 - Education Level
 - Email Address
 - First Name

Formula Bar: `1 TotalQuantity = SUM('Sales Data'[Order Quantity])`

Order Date	Order Number	Product Key	Customer Key	Territory Key	Order Line Item	Order Quantity	Index	TotalQuantity
6/3/2020	SO46718	360	12570	9	1	1	1205	84174
4/22/2020	SO46736	360	12341	9	1	1	1228	84174
5/5/2020	SO46776	360	12356	9	1	1	1267	84174
6/22/2020	SO46808	360	12347	9	1	1	1299	84174
5/11/2020	SO46826	360	12575	9	1	1	1314	84174
4/21/2020	SO47075	360	12685	9	1	1	1421	84174
5/1/2020	SO47098	360	12667	9	1	1	1445	84174
4/21/2020	SO47149	360	12669	9	1	1	1495	84174
6/4/2020	SO47212	360	12580	9	1	1	1550	84174
6/29/2020	SO47302	360	12670	9	1	1	1649	84174
8/12/2020	SO47328	360	12681	9	1	1	1669	84174
8/13/2020	SO47346	360	12585	9	1	1	1690	84174
6/12/2020	SO47744	360	12989	9	1	1	1900	84174
7/28/2020	SO47745	360	12998	9	1	1	1904	84174
8/22/2020	SO47753	360	13020	9	1	1	1912	84174

Fields List:

- Calendar Lookup
- Customer Lookup
- Product Categories Lookup
- Product Lookup
- Product Subcategories Lookup
- Returns Data
- Rolling Calendar
- Sales Data
 - Customer Key
 - Index
 - Order Date
 - Order Line Item

DAX MEASURES

- Like calculated columns, measures reference **entire tables** or **columns** (no A1-style cell references)
- Unlike calculated columns, **measures** aren't visible within tables; they can only be "seen" within a visualization like a chart or matrix (similar to a calculated field in a PivotTable)
- Measures evaluate based on **filter context**, which means they recalculate when the fields or filters around them change
- Use measures to create numerical, calculated values that can be analyzed in the "values" field of a report visual**

IMPLICIT VS. EXPLICIT MEASURES

Implicit measures are created when you drag raw numerical fields into a report visual and manually select an aggregation mode (Sum, Average, Min, Max, Count, etc.)

Explicit measures are created when you actually write a DAX formula and define a new measure that can be used within the model

Implicit measures are only accessible within the **specific visualization** in which they were created, and cannot be referenced elsewhere **Explicit measures** can be used **anywhere in the report**, and referenced by other DAX calculations to create “measure trees”.

CALCULATED COLUMNS VS. MEASURES

CALCULATED COLUMNS

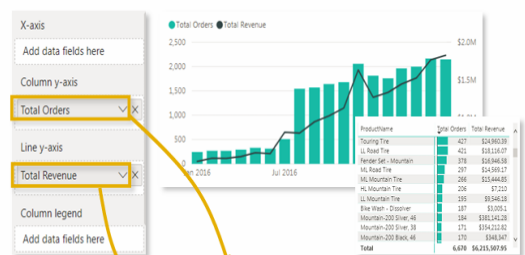
- Values are calculated based on information from each row of a table (**row context**)
- Appends static values to each row in a table and stores them in the model (*which increases file size*)
- Recalculate on data source refresh or when changes are made to component columns
- Primarily used for **filtering** data in reports

Birth Date	Marital Status	Email Address	Annual Income	Total Children	Education Level	Parent
9/3/1943	M	emma32@adventure-works.com	70000	5	Bachelors	Yes
9/24/1967	M	barry20@adventure-works.com	40000	5	High School	Yes
8/5/1945	M	martha13@adventure-works.com	70000	5	High School	Yes
6/4/1946	S	tamara16@adventure-works.com	40000	5	High School	Yes
10/10/1970	S	gerald11@adventure-works.com	130000	5	Bachelors	Yes
5/10/1945	M	alexa8@adventure-works.com	40000	5	High School	Yes
9/24/1918	M	jack53@adventure-works.com	70000	5	Graduate Degree	Yes
7/21/1959	S	ricky1@adventure-works.com	100000	5	Bachelors	Yes
1/6/1962	M	keith4@adventure-works.com	70000	5	Partial College	Yes
8/15/1962	M	lloyd11@adventure-works.com	70000	5	Bachelors	Yes
1/26/1967	S	mical11@adventure-works.com	70000	5	Bachelors	Yes
3/9/1946	M	mindy21@adventure-works.com	80000	5	Partial College	Yes
6/11/1960	M	teresa1@adventure-works.com	70000	5	Partial College	Yes

Calculated columns “live” in tables

MEASURES

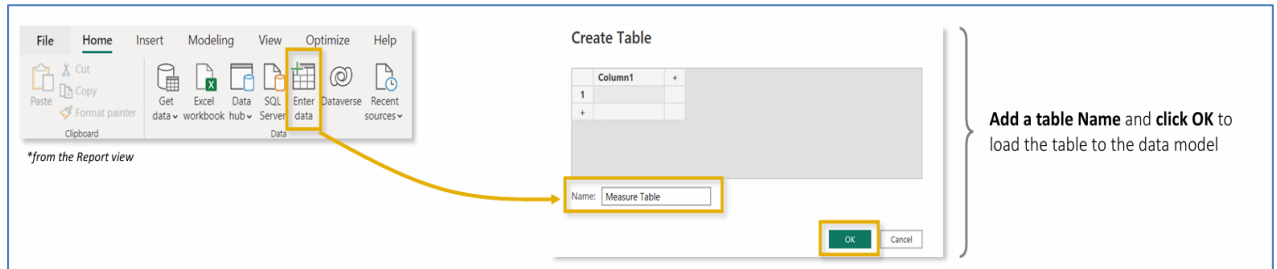
- Values are calculated based on information from any filters in the report (**filter context**)
- Does not create new data in the tables themselves (*doesn't increase file size*)
- Recalculate in response to any change to filters within the report
- Primarily used for **aggregating values** in report visuals



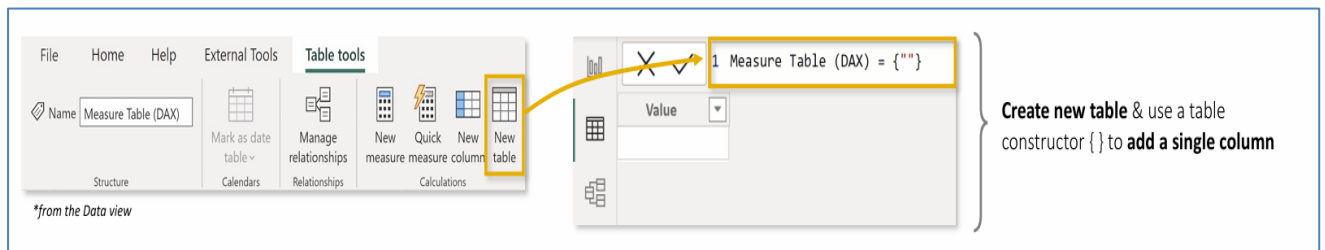
Measures “live” in visuals

MEASURE TABLES

Option 1: Enter Data into Power Query (loads the table to the data model – table is visible in Power Query)



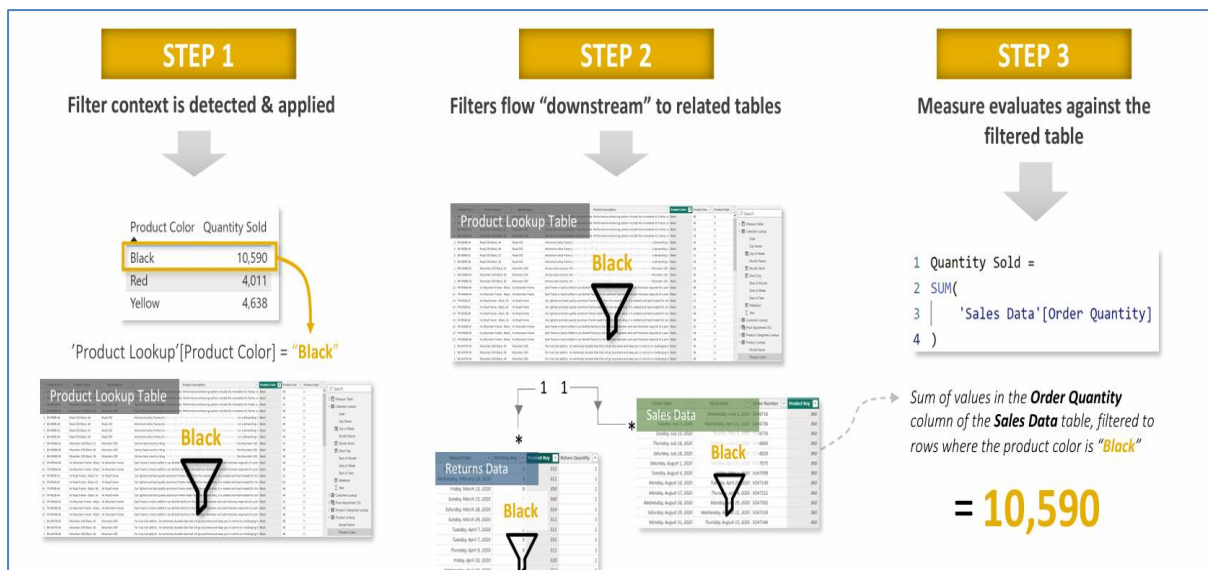
Option 2: Create a calculated table using DAX directly in the model (table is not visible in Power Query)



STEP-BY-STEP MEASURE CALCULATION

The screenshot shows a data table with two columns: 'Product Color' and 'Quantity Sold'. The rows are 'Black', 'Red', and 'Yellow'. The 'Quantity Sold' values are 10,590, 4,011, and 4,638 respectively. The value 10,590 is highlighted with a yellow box, and an arrow points from it to the right.

Product Color	Quantity Sold
Black	10,590
Red	4,011
Yellow	4,638



DAX SYNTAX

MEASURE NAME

- Measures are always surrounded by brackets (i.e. **[Total Quantity]**) when referenced in formulas, so spaces are OK

Total Quantity: = **SUM**(Transactions**[quantity]**)

Referenced
TABLE NAME

Referenced
COLUMN NAME

FUNCTION NAME

- Calculated columns don't always use functions, but measures do:
 - In a **Calculated Column**, **=Transactions[quantity]** returns the value from the quantity column in each row (since it evaluates one row at a time)
 - In a **Measure**, **=Transactions[quantity]** will return an **error** since Power BI doesn't know how to translate that as a single value – you need some sort of aggregation

This is a **"fully qualified"** column, since it's preceded by the table name.

NOTE: Table names with spaces must be surrounded by **single quotes**:

- Without a space: **Transactions[quantity]**
- With a space: **'Transactions Table'[quantity]**

PRO TIP:

Column references use fully qualified names (i.e. **'Table'[Column]**)

Measure references just use the measure name (i.e. **[Measure]**) and can be called by typing an open square bracket " ["

DAX OPERATORS

Arithmetic Operator	Meaning	Example
+	Addition	2 + 7
-	Subtraction	5 – 3
*	Multiplication	2 * 6
/	Division	4 / 2
^	Exponent	2 ^ 5

Comparison Operator	Meaning	Example
=	Equal to	[City]="Boston"
>	Greater than	[Quantity]>10
<	Less than	[Quantity]<10
>=	Greater than or equal to	[Unit Price]>=2.5
<=	Less than or equal to	[Unit Price]<=2.5
<>	Not equal to	[Country]<>"Mexico"

Text/Logical Operator	Meaning	Example
&	Concatenates two values to produce one text string	[City] & " " & [State]
&&	Create an AND condition between two logical expressions	(([State]="MA") && ([Quantity]>10))
(double pipe)	Create an OR condition between two logical expressions	(([State]="MA") ([State]="CT"))
IN	Creates a logical OR condition based on a given list (using curly brackets)	'Store Lookup'[State] IN { "MA", "CT", "NY" }

COMMON FUNCTION CATEGORIES

MATH & STATS Functions:

- SUM
- AVERAGE
- MAX/MIN
- DIVIDE
- COUNT/COUNTA
- COUNTROWS
- DISTINCTCOUNT

Iterator Functions:

- SUMX
- AVERAGEX
- MAXX/MINX
- RANKX
- COUNTX

LOGICAL Functions:

- IF
- IFERROR
- AND
- OR
- NOT
- SWITCH
- TRUE
- FALSE

TEXT Functions:

- CONCATENATE
- COMBINEVALUES
- FORMAT
- LEFT/MID/RIGHT
- UPPER/LOWER
- LEN
- SEARCH/FIND
- REPLACE
- SUBSTITUTE
- TRIM

FILTER Functions

- CALCULATE
- FILTER
- ALL
- ALLEXCEPT
- ALLSELECTED
- KEEPFILTERS
- REMOVEFILTERS
- SELECTEDVALUE

DATE & TIME Functions

- DATE
- DATEDIFF
- YEARFRAC
- YEAR/MONTH
- DAY/HOUR
- TODAY/NOW
- WEEKDAY
- WEEKNUM
- NETWORKDAYS

Time Intelligence:

- DATESYTD
- DATESMTD
- DATEADD
- DATESBETWEEN

TABLE Functions

- SUMMARIZE
- ADDCOLUMNS
- GENERATESERIES
- DISTINCT
- VALUES
- UNION
- INTERSECT
- TOPN